

HydroNex

*An AI-Integrated IoT Framework for Smart Hydroponic Farm Management
and Plant Disease Diagnosis*

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1 Problem & Market Validation

Egyptian agriculture is facing increasing pressure from water scarcity, limited arable land, and the growing impact of plant diseases on crop productivity and food security. Hydroponics offers a promising water-efficient alternative to conventional farming. However, effective hydroponic cultivation requires appropriate management of growing conditions, early identification of plant-health problems, and access to practical agricultural guidance.

At the same time, the growing hydroponics market indicates increasing commercial interest in soil-less and automated agricultural solutions. Therefore, there is a need for accessible systems that can support growers in managing both hydroponic conditions and plant health.

1.1 Water Scarcity and the Need for Hydroponic Agriculture

Egypt faces significant pressure on its freshwater resources. Its per-capita freshwater availability has fallen below 550 m³ annually, well below the internationally recognized water-scarcity threshold of 1,000 m³ per person per year [2]. In addition, less than 5% of Egypt's land is arable, while most of the country consists of desert areas [5]. These conditions create a structural challenge for conventional soil-based agriculture and increase the importance of water-efficient farming practices.

Hydroponics provides a potential response to these constraints by enabling crop production without conventional soil and with improved water-use efficiency. Consequently, soilless farming has gained increasing attention in Egypt as a potential approach to improving agricultural resource efficiency [5, 1].

The increasing market size further indicates growing interest in hydroponic agriculture. The Egyptian hydroponics market was valued at USD 102.43 million in 2024 and is projected to reach USD 159.70 million by 2030, representing a CAGR of 7.64% [8]. In addition, the automated hydroponic gardening segment was valued at USD 19.13 million in 2024 and is projected to grow at a CAGR of 10.87% through 2033 [7].

These market trends indicate increasing practical and commercial interest in hydroponic and automated agricultural solutions.

1.2 Plant Diseases and Their Impact

Water efficiency alone does not guarantee successful crop production, as plant health remains a major agricultural challenge. Plant pests and diseases are estimated to cause losses of up to 40% of global crop production each year, resulting in economic losses exceeding USD 220 billion annually [6]. Such losses make plant disease management an important concern for agricultural productivity and global food security.

For hydroponic growers, timely identification of plant-health problems is particularly important because many hydroponic systems rely on shared and recirculating nutrient solutions. Consequently, a pathogen introduced into a system may potentially affect multiple plants if the problem is not identified and addressed promptly.

1.3 Target Users and Their Challenges

The proposed solution primarily targets hydroponic growers, particularly small-scale and beginner growers who may have limited access to agricultural specialists or diagnostic services. Commercial growers and greenhouse operators may also benefit from continuous monitoring and early identification of plant-health problems.

The main challenges faced by these users include:

- Difficulty identifying plant diseases at an early stage.
- Dependence on manual visual inspection and personal experience.

- Limited access to agricultural specialists and diagnostic services.
- The need for continuous monitoring of hydroponic growing conditions.
- Difficulty obtaining immediate and understandable guidance when a problem occurs.

1.4 Existing Challenges and Market Need

Despite the increasing interest in hydroponics, adoption can still be affected by technical complexity, initial costs, and the need for specialized knowledge. These challenges may be particularly significant for beginners and small-scale growers who do not have continuous access to agricultural experts.

Hydroponic systems commonly require monitoring of parameters such as pH, electrical conductivity (EC), temperature, humidity, water level, and other environmental conditions. Although these measurements are essential for maintaining suitable growing conditions, they do not directly identify whether a plant is affected by a disease. Plant-health assessment may therefore still depend on manual visual inspection.

In addition, agricultural support may be fragmented across different sources. A grower may need separate resources for system monitoring, plant-disease identification, and agricultural advice. This fragmentation can create an additional knowledge barrier, particularly for inexperienced users.

1.5 Market Gap

Based on the identified challenges, the main gap addressed by this project can be summarized in three areas:

1. **Accessibility Gap:** Hydroponic cultivation can involve technical and financial barriers that make adoption more difficult for beginners and small-scale growers.
2. **Plant-Health Monitoring Gap:** Existing monitoring approaches can track environmental conditions but do not necessarily provide direct and early identification of plant diseases.
3. **Knowledge Gap:** Growers may need to consult multiple sources or specialists to understand plant symptoms and determine appropriate actions.

1.6 Proposed Value Proposition

This project addresses these gaps through an integrated hydroponic support platform that combines three complementary functions: **plant disease detection, hydroponic system monitoring, and an intelligent chatbot.**

The platform is designed to help users identify potential plant diseases from images, monitor important growing conditions, and obtain accessible guidance when problems occur. By integrating these functions within a single platform, the project aims to provide a more accessible approach to hydroponic crop management and support earlier identification and response to potential plant-health problems.

2 AI Approach & Justification

2.1 Overview

This section explains the AI functionality of the HydroSentry platform, why each component was chosen, and how it addresses the core problem described in Section 1: giving the farmer a single, explainable source of truth instead of manually cross-checking sensors, plant photos, and past experience. It first presents the complete pipeline end-to-end, then walks through each stage in detail with example inputs and outputs.

2.2 Complete AI Pipeline

One full monitoring cycle, from the raw camera image to the recommendation shown on the farmer's dashboard, is illustrated in Figure 1.

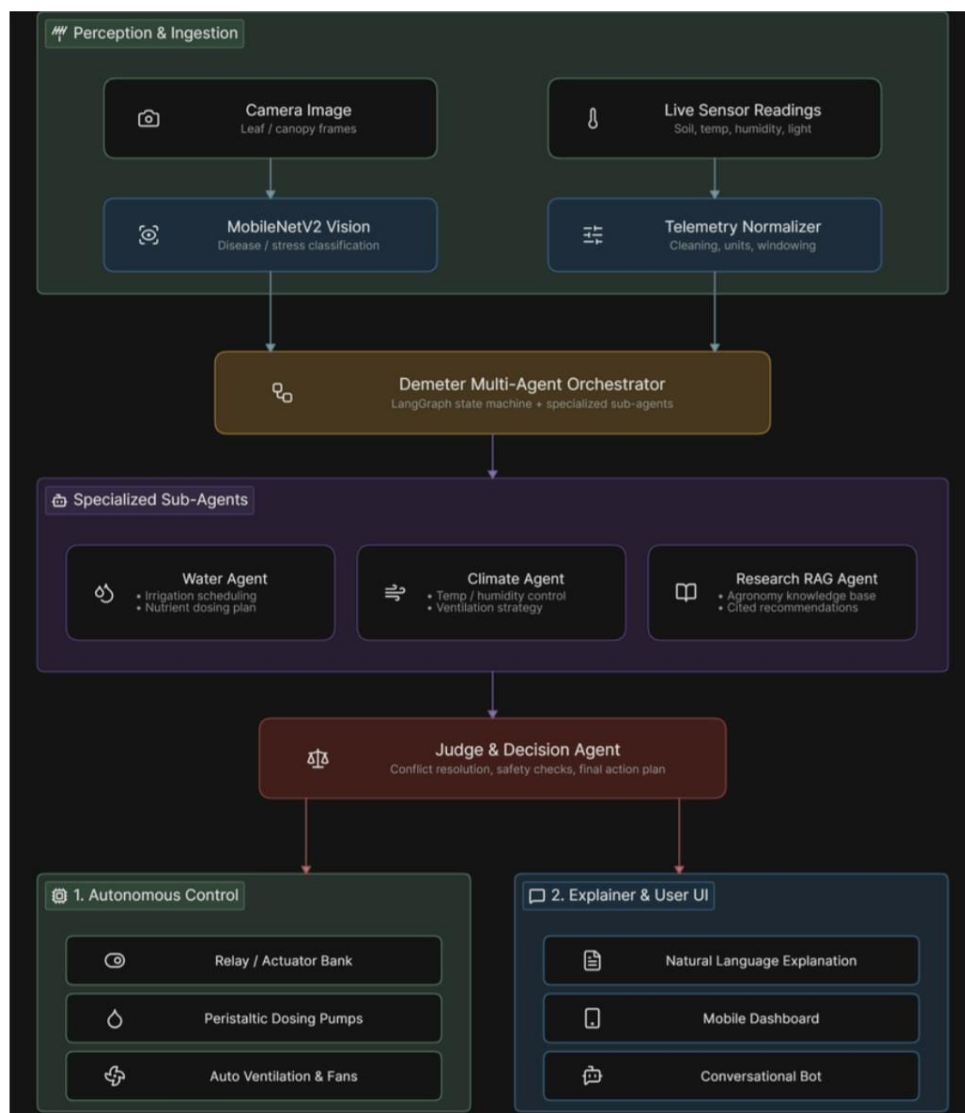


Figure 1: Complete AI pipeline: sensor data and historical RAG matches are fetched in parallel with disease classification, then fused inside the multi-agent system before a final, explained recommendation is produced.

2.3 Pipeline Stages

2.3.1 Data Collection — Continuous Multi-Source Ingestion

Before any analysis takes place, the system continuously collects data from four sources so that every downstream decision is grounded in complete, up-to-date context:

- **Environmental sensors:** temperature, humidity, CO₂.
- **Water sensors:** pH, EC, water temperature.
- **Plant images:** captured by a camera.
- **Historical farm records:** stored in the vector database.

2.3.2 Plant Disease Detection — Computer Vision on Captured Leaf Images

A captured plant image is sent to the **MobileNetV2 Plant Disease Identification** model.

Purpose:

- Detect plant diseases from leaf images.
- Classify plant conditions into disease categories.
- Generate confidence scores for predictions.

Input	Leaf image
Output	Disease name, confidence score

Example output

Input image → Tomato leaf

Prediction: **Tomato Early Blight**

Confidence: **96%**

2.3.3 Sensor Data Processing — Normalizing the Latest Readings

The latest sensor readings are collected and normalized before being used by any downstream model or agent.

Example --- normalized reading

pH: 5.1

EC: 0.9

Temperature: 29°C

Humidity: 75%

2.3.4 Historical Case Retrieval (RAG) — Learning from Past Farm Experience

The current situation is converted into embeddings and searched against the **Qdrant Vector Database**.

Technologies used: Qdrant; embedding model (FastEmbed / Sentence Transformers).

The system retrieves:

- Similar environmental conditions.
- Similar disease cases.
- Previous successful treatments.

Purpose: learn from previous farm experiences instead of analyzing the situation from scratch.

2.3.5 Multi-Agent Analysis — Powered by Azure OpenAI GPT-4.1

The retrieved historical context and live sensor data are passed to the **Demeter** multi-agent system. Three specialized agents analyze the situation in parallel, each producing a focused recommendation:

Water Agent	Atmospheric Agent	Research Agent
Analyzes: pH, EC, water quality	Analyzes: temperature, humidity, air conditions	Analyzes: retrieved historical cases, scientific reasoning, disease context
Produces: water-related recommendations	Produces: climate-related recommendations	Produces: evidence-based recommendations

2.3.6 Decision Making — Judge Agent, powered by Azure OpenAI GPT-4.1

The Judge Agent receives the outputs from all three specialist agents and determines the current system status, risk level, and best corrective action.

Example output

Risk level: **Medium**

Recommended action: reduce humidity and apply treatment protocol for Early Blight.

2.3.7 Explainable AI Layer — Explainer Agent, powered by Azure OpenAI GPT-4.1

The Explainer Agent generates a human-readable explanation describing the detected disease, sensor observations, retrieved historical evidence, and the reasoning behind the recommendation — so the farmer never receives a black-box answer.

Example explanation

Observation: humidity reached 75%.

Disease detection: **Tomato Early Blight detected with 96% confidence.**

Historical evidence: similar cases showed successful recovery after reducing humidity below 65%.

Conclusion: lower humidity and begin disease treatment.

2.3.8 Final Recommendation — What the Farmer Actually Receives

Farmer-facing summary

Detected disease: **Tomato Early Blight**

Confidence: 96%

Current status: **Moderate risk**

Recommended actions:

- Reduce humidity
- Adjust irrigation schedule
- Apply treatment protocol
- Continue monitoring for 48 hours

2.3.9 Autonomous Actuation — Closed-Loop Execution without Human Delay

The Judge Agent evaluates system criticality and routes automated execution commands directly to local microcontrollers to maintain chemical and environmental equilibrium.

Purpose:

- Execute time-sensitive corrections autonomously (pH drift, nutrient depletion, heat stress).
- Eliminate reliance on immediate manual intervention.
- Separate routine automated balancing from high-level farmer interventions.

Input: final decision and corrective actions from the Judge Agent; real-time sensor baselines.

Output: hardware actuation signals (PWM / relay triggers); execution logs sent to the cloud database.

Example output

```
Target actuator: Peristaltic Pump 01 (pH Down Buffer)
Action type: PWM pulse injection (duration: 2.5s)
Secondary action: exhaust fan active (target humidity < 65%)
Status: executed autonomously in 120ms
```

2.3.10 Conversational Assistant — Interactive Agronomic Dialogue & Farmer Q&A

A conversational RAG assistant powered by LLMs interfaces directly with the live telemetry state, computer vision diagnosis, and historical records to answer farmer questions via voice or text.

Purpose:

- Allow natural-language inquiries regarding real-time farm status and crop health.
- Provide transparent agronomic reasoning behind every autonomous action taken.
- Support manual vocal overrides and custom operational queries.

Input: farmer verbal/text query; live normalized sensor telemetry and MobileNetV2 disease report.

Output: grounded natural-language explanation and interactive guidance.

Example interaction

```
Farmer query: ``Why did the acid dosing pump turn on 5
minutes ago?``
AI assistant: ``The solution pH drifted to 5.1, impairing
nutrient availability. The system injected 2.5s of buffer
to stabilize the reservoir back to the optimal 5.8 target.``
```

2.4 AI Technologies Used

Component	Technology
Disease Detection	MobileNetV2 Plant Disease Identification
Historical Memory	Qdrant Vector Database
Embeddings	FastEmbed / Sentence Transformers
Multi-Agent Framework	Demeter
Large Language Model	Azure OpenAI GPT-4.1 / Gemini
Agent Orchestration	LangGraph
AI Framework	LangChain
Autonomous Actuation	Closed-Loop Actuator Engine
Interactive Assistant	Conversational Multimodal RAG

2.5 Why This AI Approach?

- MobileNetV2 provides fast and accurate plant disease detection from leaf images.
- Sensor analytics continuously monitors hydroponic conditions.
- Qdrant RAG retrieves relevant historical cases and successful treatments.
- The Demeter multi-agent architecture enables specialized analysis through Water, Atmospheric, Research, Judge, and Explainer agents.
- Azure OpenAI GPT-4.1 performs advanced reasoning, decision-making, and explanation generation.
- Explainable AI ensures transparency and user trust.

This architecture combines **Computer Vision**, **Retrieval-Augmented Generation (RAG)**, **Multi-Agent Reasoning**, and **Large Language Models** to deliver intelligent, explainable, and actionable recommendations for hydroponic farming.

3 Technical Architecture

This section outlines the system design, pipeline flow, backend stack, and the integration of AI components for the Smart Hydroponic Farm Management System. It ensures full compatibility with the advanced multi-agent AI approach and autonomous control mechanisms.

3.1 System Design

The architecture is built upon a robust, scalable, four-layer microservices-oriented design:

- **IoT Sensing & Actuation Layer:** Microcontrollers (e.g., ESP32) interfaced with environmental sensors (Temperature, Humidity, pH, EC, Water Level, Light, CO₂) and connected to relay/actuator banks (Peristaltic Dosing Pumps, Auto Ventilation).
- **Data & Connectivity Layer:** Wi-Fi enabled transmission pushing telemetry data to the cloud via REST APIs, and receiving hardware actuation signals (PWM/Relay Triggers).
- **Backend & AI Layer (Cloud):** A centralized orchestration backend handling business logic, data persistence, and seamless communication with isolated AI microservices and the Demeter Multi-Agent Orchestrator.
- **Application Layer:** A cross-platform mobile dashboard providing real-time monitoring, natural language explanations, and a conversational bot interface.

3.2 Backend Stack & Infrastructure

- **Core Framework:** C# ASP.NET Core implementing Clean Architecture and the CQRS pattern (via MediatR) to ensure a highly maintainable, loosely coupled, and testable codebase capable of handling continuous multi-source ingestion.
- **Relational Database:** SQL Server managed through Entity Framework (EF) Core, storing historical time-series sensor data, user profiles, and execution logs.
- **Vector Database (RAG):** Qdrant is utilized to store and retrieve historical farm records, previous successful treatments, and the agronomic knowledge base using embeddings (FastEmbed/Sentence Transformers).
- **Real-time Communication:** SignalR is utilized to establish WebSockets, pushing live sensor updates, instant AI alerts, and conversational bot responses directly to the mobile application.

3.3 Pipeline Flow

The data life-cycle, from perception to autonomous actuation and farmer notification, operates as follows:

1. **Perception & Ingestion:** The ESP32 continuously reads sensors and sends payloads to the ASP.NET Core API. Simultaneously, plant images are captured and sent to the Vision API.
2. **Processing & Normalization:** The backend validates the incoming data, stores it in SQL Server, and passes it through a Telemetry Normalizer (cleaning limits, windowing) before forwarding it to the AI layer.
3. **Multi-Agent Analysis:** The normalized telemetry, vision classification (MobileNetV2), and retrieved historical context (Qdrant RAG) are fed into the Demeter Multi-Agent Orchestrator (powered by LangGraph and Azure OpenAI GPT-4.1). Specialized sub-agents (Water, Climate, Research) analyze the situation in parallel.

4. **Decision & Actuation:** The Judge & Decision Agent synthesizes the sub-agent outputs, resolves conflicts, and determines the corrective action. If critical, the backend routes autonomous execution commands (e.g., PWM pulse to a dosing pump) directly to the IoT Actuation Layer.
5. **Explainability & Alerting:** The Explainer Agent generates a human-readable summary of the actions taken and the reasoning behind them, which the backend broadcasts via SignalR to the Application Layer.

3.4 How AI Fits In (Integration Strategy)

To optimize performance and maintain separation of concerns, the AI components are decoupled from the main operational backend:

- **Microservices Architecture:** The AI models (Plant Vision, Embedding Services) and the Multi-Agent Framework are hosted as independent Python-based REST APIs.
- **Computer Vision Pipeline:** When a user captures a plant image, the ASP.NET Core API acts as a gateway, forwarding the image to the MobileNetV2 Vision service, which returns the disease classification and confidence score.
- **Conversational Assistant:** A conversational RAG assistant powered by LLMs interfaces directly with the live telemetry state. When a farmer asks a query (e.g., "Why did the acid dosing pump turn on?"), the backend aggregates the latest telemetry and execution logs, injects them into the prompt, and queries the LLM for a grounded, context-aware response.

4 Business Feasibility

AI-Powered Hydroponic Farm Management System — Egyptian Market

4.1 Target Audience

The proposed platform targets several user segments within the hydroponic agriculture ecosystem:

Segment	Profile	Why They Need This
Small and medium-scale growers	Hydroponic farming operations with limited technical resources	Need continuous monitoring and early identification of plant-health problems without relying entirely on manual inspection
Urban and rooftop growers	Individuals operating small hydroponic systems in urban environments	Need accessible monitoring and guidance to reduce the technical barrier associated with hydroponic cultivation
University and research setups	Agricultural education programs and research greenhouses	Need structured sensor data, historical records, and plant-health tracking for experimentation and research

4.2 Market Relevance

The Egyptian hydroponics market was valued at approximately USD 102.43 million in 2024 and is projected to reach USD 159.70 million by 2030, representing a CAGR of 7.64% [8].

The automated hydroponic gardening segment was valued at USD 19.13 million in 2024 and is projected to grow at a CAGR of 10.87% through 2033 [7].

For consistency, the following table presents the reported market values and their corresponding sources:

Metric	Reported Figure	Source
Egypt hydroponics market (2024)	USD 102.43 Million	[8]
Egypt hydroponics market (2030F)	USD 159.70 Million	[8]
Automated hydroponic gardening segment (2024)	USD 19.13 Million	[7]
Automated hydroponic gardening segment CAGR	10.87% through 2033	[7]

The combination of water scarcity and growing market interest supports the relevance of water-efficient and technologically supported agricultural solutions in Egypt.

4.3 Timing — Why Now

Beyond the overall size of the market, several converging factors indicate that this is a particularly favorable period for launching an AI-integrated hydroponic management platform in Egypt, rather than a generic long-term opportunity.

- **The automated segment is growing faster than the market around it.** The broader Egyptian hydroponics market is expanding at a 7.64% CAGR, while the automated hydroponic gardening segment specifically is forecast to grow at a steeper 10.87% CAGR through 2033 [8, 7]. This gap suggests

that growers are not simply adopting hydroponics, but increasingly favoring automated, technology-assisted systems over manual ones — a shift a monitoring-and-diagnosis platform is positioned to serve directly.

- **National policy is actively pushing water-efficient agriculture on the same timeline.** Egypt’s 2025/2026 Socio-Economic Development Plan and Vision 2030 explicitly target raising the share of agricultural land under water-efficient practices from roughly 30% in 2023/2024 to 100% by 2030 [3]. A platform that helps growers operate hydroponic systems more effectively is directly aligned with, and stands to benefit from, this active policy push rather than working against it.
- **The underlying AI infrastructure only recently became practical to build on.** Egypt launched the second edition of its National AI Strategy in 2026, and government-backed efforts have already produced a homegrown large language model for agriculture and other sectors [4]. Combined with the continued drop in cost and improved reliability of commercial LLM APIs and mobile-friendly vision models (see Section 2), the technical building blocks this platform depends on — affordable disease classification and grounded conversational assistance — are now mature and inexpensive enough for a graduation-project team to deploy them credibly, which was not realistically true even two to three years earlier.
- **The competitive gap is still open.** As the following section shows, existing Egyptian players offer either hardware/consulting or fully manual advisory services, but none yet combine IoT monitoring, AI-based disease detection, and a conversational assistant in one product. Entering while this integrated approach is still uncommon locally is more advantageous than waiting until an incumbent closes the gap.

4.4 Competitive Landscape

Existing companies in the Egyptian hydroponic and controlled-agriculture market demonstrate that there is an established interest in hydroponic systems and agricultural automation. The proposed project differentiates itself by integrating monitoring, plant-health assessment, and intelligent guidance within a single platform.

Competitor	General Offering	Potential Gap
Green Tower Egypt	Hydroponic and aeroponic growing systems	Focus primarily on physical growing systems rather than an integrated AI-based monitoring platform
Alex Hydroponics	Hydroponic systems and related agricultural services	Does not primarily focus on an integrated AI-based monitoring and plant-health platform
Plug’n’Grow	Hydroponic installation and consulting services	The proposed system additionally focuses on automated monitoring and AI-assisted plant-health assessment
AlGhanim Agriculture Group	Controlled-environment and automated greenhouse solutions	The proposed system targets a more integrated self-service AI and IoT management approach

The proposed differentiation is therefore based on the integration of **IoT monitoring, AI-based plant disease detection, predictive analysis, and an intelligent conversational assistant** within one platform.

4.5 Viability & Execution Scope

The proposed MVP relies primarily on existing and proven technologies rather than requiring the development of entirely new hardware or AI research. The system can be developed using pretrained or fine-tuned disease-detection models, existing language-model APIs, commercially available IoT sensors, and standard microcontrollers and actuators.

This approach makes the proposed system technically feasible within a graduation-project or competition development timeline while allowing the AI and software components to remain extensible for future development.

4.6 Risk Mitigation

Risk	Mitigation
Higher upfront hardware cost compared with traditional farming	Position the AI and software components as an additional layer that can operate with commonly available hydroponic hardware rather than requiring a completely new farming system
Currency exposure for imported components or USD-denominated cloud services	Prefer locally available hardware where possible and consider pricing the software component in EGP
Limited availability of Egypt-specific pricing benchmarks for an integrated AI and IoT hydroponic platform	Defer final pricing decisions until additional market validation and customer research are conducted

Pricing Assumption: Exact product pricing is left open for a later phase because a reliable Egypt-specific pricing benchmark for this exact product category has not yet been established. This approach avoids introducing unsupported pricing assumptions.

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